# Jesse Zhang

Postdoc focused on Post-Deployment Robot Learning at UW with Abhishek Gupta and Dieter Fox
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## Education

**University of Southern California**, Los Angeles, CA — 2020 - 2025
*Ph.D.* in Computer Science (Advisors: Erdem Biyik, Joseph Lim, Jesse Thomason) — GPA: 4.00/4.00

**UC Berkeley**, Berkeley, CA — 2016 - 2020
*B.A.* in Computer Science (Honors in CS, Graduated w/ Highest Distinction) — GPA: 3.96/4.00

## In Submission

[S3] Ge Yan*, Jinghao Liu*, Yuzhi Fan*, Lei Cai, Minwen Liao, **Jesse Zhang**†, and Dieter Fox†. "Flex-$\pi$: A Multi-Stream World-Action Model with Compute Flexibility", *arXiv preprint arXiv:2608.10860*, 2026

[S2] Marius Memmel, Shawn Yin, Urvi Rutia, Gavin Angeloff, Eldon Li, Ankit Goyal, **Jesse Zhang**, Dieter Fox, Fabio Ramos, Anqi Li†, and Abhishek Gupta†. "D&A: Decoupling Description and Assessment Enables VLMs as Zero-Shot Robotic Reward Models", *In Submission*, 2026

[S1] Skand Peri*, Matthew Hong*, Chaoxiang Zhang, Ross Worobel, **Jesse Zhang**†, and Abhishek Gupta†. "VETO: Reward-Supervised World Models as Action Verifiers for Sample-Efficient RL", *In Submission*, 2026

## Conference Papers

[C19] Sarvesh Patil*, Mitsuhiko Nakamoto*, Manan Agarwal, Shashwat Saxena, **Jesse Zhang**, Giri Anantharaman, Cleah Winston, Chaoyi Pan, Douglas Chen, Nai-Chieh Huang, Zeynep Temel, Oliver Kroemer, Abhishek Gupta, Sergey Levine, Hongkai Dai*, Paarth Shah*, and Max Simchowitz*. "OGPO: Sample Efficient Full-Finetuning of Generative Control Policies", *ICML*, 2026

[C18] Matthew Hong, **Jesse Zhang**, Anusha Nagabandi, and Abhishek Gupta. "Diffusion Timestep-Modulated Pre-training Enables Exploration for Efficient Policy Fine-tuning", *RSS, World Modeling Workshop, Oral at RL4IL ICRA Workshop*, 2026

[C17] Anthony Liang*, Yigit Korkmaz*, Jiahui Zhang, Minyoung Hwang, Abrar Anwar, Sidhant Kaushik, Aditya Shah, Alex S Huang, Luke Zettlemoyer, Dieter Fox, Yu Xiang, Anqi Li, Andreea Bobu, Abhishek Gupta, Stephen Tu†, Erdem Biyik†, and **Jesse Zhang**†. "Robometer: Scaling General-Purpose Robotic Reward Models via Trajectory Comparisons", *RSS, Oral at RL4IL ICRA Workshop*, 2026

[C16] Matthew Hong*, Anthony Liang*, Kevin J. Kim, Harshitha Belagavi Rajaprakash, Jesse Thomason†, Erdem Biyik†, and **Jesse Zhang**†. "HAND Me The Data: Fast Robot Adaptation via Hand Path Retrieval", *ICRA*, 2026

[C15] **Jesse Zhang***, Marius Memmel*, Kevin Kim, Dieter Fox, Jesse Thomason, Fabio Ramos, Erdem Biyik, Abhishek Gupta†, and Anqi Li†. "PEEK: Guiding and Minimal Image Representations for Zero-Shot Generalization of Robot Manipulation Policies", *ICRA*, 2026

[C14] Jiahui Zhang*, Yusen Luo*, Abrar Anwar*, Sumedh Anand Sontakke, Joseph J. Lim, Jesse Thomason, Erdem Biyik, and **Jesse Zhang**. "ReWiND: Language-Guided Rewards Teach Robot Policies without New Demonstrations", ***Oral at CoRL (top 4.3%)***, 2025

[C13] Yi Li*, Yuquan Deng*, **Jesse Zhang***, Joel Jang, Marius Memmel, Caelan Reed Garrett, Fabio Ramos, Dieter Fox, Anqi Li†, Abhishek Gupta†, and Ankit Goyal†. "HAMSTER: Hierarchical Action Models for Open-World Robot Manipulation", *ICLR*, 2025

[C12] **Jesse Zhang**, Minho Heo, Zuxin Liu, Erdem Biyik, Joseph J Lim, Yao Liu, and Rasool Fakoor. "EXTRACT: Efficient Policy Learning by Extracting Transferable Robot Skills from Offline Data", *CoRL*, 2024

[C11] Yufei Wang, Zhanyi Sun, **Jesse Zhang**, Zhou Xian, Erdem Biyik, David Held, and Zackory Erickson. "RL-VLM-F: Reinforcement Learning from Vision Language Foundation Model Feedback", *ICML*, 2024

[C10] **Jesse Zhang**, Karl Pertsch, Jiahui Zhang, and Joseph J Lim. "SPRINT: Scalable Semantic Policy Pre-training via Language Instruction Relabeling", *ICRA 2024. LangRob Workshop CoRL 2022 Spotlight*,

[C9] Zuxin Liu, **Jesse Zhang**, Kavosh Asadi, Yao Liu, Ding Zhao, Shoham Sabach, and Rasool Fakoor. "TAIL: Task-Specific Adapters for Imitation Learning", *ICLR*, 2024

[C8] Sumedh Anand Sontakke, **Jesse Zhang**, Séb Arnold, Karl Pertsch, Erdem Biyik, Dorsa Sadigh, Chelsea Finn, and Laurent Itti. "RoboCLIP: One Demonstration is Enough to Learn Robot Policies", *NeurIPS*, 2023

[C7] **Jesse Zhang**, Jiahui Zhang, Karl Pertsch, Ziyi Liu, Xiang Ren, Minsuk Chang, Shao-Hua Sun, and Joseph J Lim. "Bootstrap Your Own Skills: Learning to Solve New Tasks with Large Language Model Guidance", ***Oral at CoRL (top 6.6%)***. *Spotlight at RSS Articulate Robots Workshop*, 2023

[C6] Dweep Trivedi*, **Jesse Zhang***, Shao-Hua Sun, and Joseph J. Lim. "Learning to Synthesize Programs as Interpretable and Generalizable Policies", *NeurIPS*, 2021

[C5] **Jesse Zhang***, Haonan Yu*, and Wei Xu. "Hierarchical Reinforcement Learning by Discovering Intrinsic Options", *ICLR*, 2021

[C4] Avi Singh, Albert Yu, Jonathan Yang, **Jesse Zhang**, Aviral Kumar, and Sergey Levine. "COG: Connecting New Skills to Past Experience with Offline Reinforcement Learning", *CoRL*, 2020

[C3] **Jesse Zhang**, Brian Cheung, Chelsea Finn, Sergey Levine, and Dinesh Jayaraman. "Cautious Adaptation For Reinforcement Learning in Safety-Critical Settings", *ICML*, 2020

[C2] **Jesse Zhang**, Jack Sullivan, Vasudev Venkatesh PB, Kyle Tse, Andy Yan, John Leyden, Kalyanaraman Shankari, and Randy H Katz. "TripAware: Emotional and Informational Approaches to Encourage Sustainable Transportation via Mobile Applications", *ACM BuildSys*, 2019

[C1] Brian Yang, **Jesse Zhang**, Vitchyr Pong, Sergey Levine, and Dinesh Jayaraman. "REPLAB: A Reproducible Low-Cost Arm Benchmark for Robotic Learning", *ICRA*, 2019

## Journal Papers

[J2] Yongshuai Liu, Jiangyi Xia, Ziwen Kan, **Jesse Zhang**, Sheela Toprani, James B. Brewer, Marta Kutas, Xin Liu, and John Olichney. "Unveiling Early Signs of Preclinical Alzheimer's Disease Through ERP Analysis with Weighted Visibility Graphs and Ensemble Learning", *Bioengineering*, 2025

[J1] **Jesse Zhang**, Jiangyi Xia, Xin Liu, and John Olichney. "Machine Learning on Visibility Graph Features Discriminates the Cognitive Event-Related Potentials of Patients with Early Alzheimer's Disease from Healthy Aging", *Brain Sciences*, 2023

## Preprints and Workshop Papers

[P7] Ryan Lindeborg, Richard Peng, Ayush Jain, **Jesse Zhang**, Abrar Anwar, and Jesse Thomason. "Anticipating Unintended Robot Behaviors with Side Effect Critics", *Spotlight at Workshop on Rethinking What It Means to be "Safe" for Generalist Robots*, 2026

[P6] Anthony Liang*, Yigit Korkmaz*, Jiahui Zhang*, **Jesse Zhang***, Abrar Anwar, Sidhant Kaushik, Yufei Wang, Yu Xiang, David Held, Dieter Fox, Abhishek Gupta, Stephen Tu†, and Erdem Biyik†. "SPUR: Scaling Reward Learning from Human Demonstrations", *CoRL 2025 Workshop on Evaluation and Deployment Across the Robot Learning Lifecycle*, 2025

[P5] Taewook Nam, Juyong Lee, **Jesse Zhang**, Sung Ju Hwang, Joseph J Lim, and Karl Pertsch. "LiFT: Unsupervised Reinforcement Learning with Foundation Models as Teachers", *2nd Workshop on Agent Learning in Open-Endedness (ALOE) at NeurIPS 2023*, 2023

[P4] Linghan Zhong, Ryan Lindeborg, **Jesse Zhang**, Joseph J Lim, and Shao-Hua Sun. "Hierarchical Neural Program Synthesis", *ArXiv Preprint*, 2023

[P3] **Jesse Zhang***, Karl Pertsch*, Jiefan Yang, and Joseph J Lim. "Minimum Description Length Skills for Accelerated Reinforcement Learning", *ICLR 2021 Self-Supervision for Reinforcement Learning Workshop*, 2021

[P2] Kalyanaraman Shankari, Jonathan Fuerst, Mauricio Fadel Argerich, Eleftherios Avramidis, and **Jesse Zhang**. "MobilityNet: Towards A Public Dataset For Multi-Modal Mobility Research", *ICLR Climate Change AI Workshop 2020*, 2020

[P1] Daiyaan Arfeen* and **Jesse Zhang***. "Unsupervised Projection Networks for Generative Adversarial Networks", *ICCV 2019 Sensing, Understanding, and Synthesizing Humans Workshop*, 2019

## Honors and Awards

- RSS Pioneer, 30 selected out of 186 applicants 2026
- Best Paper Award (ReWiND) at OOD Workshop RSS 2025
- Best Paper Nomination (ReWiND) at RoboReps Workshop RSS 2025
- Qualcomm Innovation Fellowship Finalist 2024
- Best Paper Runner-up, CoRL LangRob Workshop 2022
- Highlighted Reviewer Award (top 8%), ICLR 2022
- Distinguished Reviewer Award (top 8%), NeurIPS 2021
- Travel Award, ICLR 2020
- Honors in Computer Science, UC Berkeley 2020

## Experience

*Contractor* April 2026 - Current
Toyota Research Institute through HireArt, Los Altos, CA

*Research Collaborator* January 2026 - Current
Allen Institute for AI, Seattle, WA

*Postdoctoral Researcher* July 2025 - Current
University of Washington, Seattle, WA

*Research Scientist Intern* May 2024 - March 2025
NVIDIA Seattle Robotics Lab, Seattle, WA

*Applied Scientist Intern* June 2023 - November 2023
Amazon Lablets, Santa Clara, CA

*Research Intern* February 2021 - August 2021
NAVER CLOVA AI Research, Seongnam, Korea

*Research Intern* January 2020 - August 2020
Horizon Robotics, Cupertino, CA

## Services

**Workshop Organizing**

- Roadmap to Sample-Efficient Real-World Reinforcement Learning at CoRL 2026 (Lead Organizer)
- Eval & Deploy at CoRL 2025

**Reviewer**

- RL Conference (Senior/Holistic Reviewer), RA-L, ICRA, NeurIPS, ICML, ICLR, CoRL, HRI 2025, TMLR 2022-2024, UIST 2024, CHI 2024, IEEE ITSC 2019
- Area Chair: Pre-train to Adapt workshop at CoRL 2026

**Mentoring and Outreach**

- **UW PAMS Mentoring**: Mentoring PhD applicants to UW to assist with application prep and general questions about PhD life.
- **UW ML Reading Program**: 1:1 meetings reading ML research papers with undergrads.
- **USC UGrad Mentoring Program**: 1:1 meetings to forge paths for getting involved in research.
- **UROS Robotics Symposium**: Organizer of PhD-student-led robotics symposium at USC.
- **CURVE**: Mentored USC undergrads through fellowship program.
- **Google x USC AI Community Project**: Helped undergrads design AI education outreach programs for underrepresented students in K-12 schools and Los Angeles community events. .

## Teaching

*Graduate Student Instructor*, USC 2020-2025
CSCI-360 Intro to AI; CSCI-566 Deep Learning

- TA'd 5 times in Intro to AI, Deep Learning. Median TA rating 5/5, mean 4.6/5.

*Undergraduate Student Instructor*, UC Berkeley Fall 2019
CS 188: Intro to AI (Anca Dragan)

- Lead a discussion section and held office hours — received 4.75/5 rating, 0.42 above dept avg

*Course Reader*, UC Berkeley Spring 2019
CS 170: Algorithms/Intro to CS Theory (Lucas Trevisan and Prasad Raghavendra)

- Held office hours + volunteered to write problems for and help run extra sections on difficult material.

## Research Mentoring

**PhD Students**

- Sriyash Poddar (UW) Current
- Harine Ravichandran (UW) Current
- Arjun Guru (UW) Current
- Cornelius Braun (Ai2) Current
- Ge Yan (UW) Current
- Marius Memmel (UW) ICRA 2026, Current
- Abrar Anwar (USC) CoRL 2025, RSS 2026

- Anthony Liang (USC) — ICRA 2026, RSS 2026
- Yigit Korkmaz (USC) — RSS 2026

**Undergraduate, Masters, and Visiting Researchers**

- Hyeonho Oh (USC) — Current
- Yixin Zhang (UW) — Current
- Yuzhi Fan (UW) — Current
- Jason Lee (Ai2, UW) — Current
- Aditya Shah (UW) — RSS 2026
- Cleah Winston (UW) — ICML 2026
- Matthew Hong (USC + UW) — ICRA 2026, RSS 2026
- Yusen Luo (USC) — CoRL 2025
- Ryan Lindeborg (USC) — 2023 Preprint, RSS Workshop 2026
- Jiahui Zhang (USC) — CoRL 2023, ICRA 2024, CoRL 2025
- Jiankun (Richard) Peng (USC) — RSS Workshop 2026
- Kevin Kim (USC) — CoRL 2025, ICRA 2026
- Jiefan Yang (USC) — ICLR Workshop 2021
- Sarthak Bhagat (USC) — 2020-2021
- Dweep Trivedi (USC) — NeurIPS 2021

## Talks and Panels

**"Making Robot Data Count: Toward Robots That Keep Learning After Deployment"**

- UPenn Grasp SFI Seminar — October 2026
- Brown, BU, Columbia, Harvard, JHU, MIT, Northeastern, Princeton, UofT — October 2026

**Invited Panel Discussion**

- RL4VLA Workshop at RSS — July 2026
- Post-Training for Robotics Foundation Models Workshop at RSS — July 2026
- SemRob Workshop at RSS — July 2026
- RLxF Workshop at ICML — July 2026

**"Learn Less, Borrow More: Generalization doesn't have to be re-learned"**

- SemRob Workshop at RSS — July 2026

**"Beyond Human Feedback: Synthetic Preferences for Scalable Robot RL"**

- RLxF Workshop at ICML — July 2026

**"Duty Free: Robot Learning with Minimal Human Supervision"**

- Waymo (Host: Matthew O' Kelly) — June 2026
- UT Dallas (Host: Prof. Yu Xiang) — April 2026

**"Robometer"**

- NVIDIA Seattle Robotics Lab (Host: Ankur Handa) March 2026

**"Scaling Robotics: From Generalists to Specialists"**
- UC San Diego (Host: Prof. Xiaolong Wang) August 2025

**"Hierarchical VLAs for Robot Manipulation"**
- OpenAI Robotics Reading Group March 2025

**"Scalable robot adaptation with large pre-trained models"**
- Google Deepmind Tech Talk Series (Host: Jie Tan) February 2025
- Yonsei University RLLab (Prof. Youngwoon Lee) January 2025
- NTU Robot Learning Lab (Prof. Shao-Hua Sun) December 2024
- 44th Southern California Control Workshop November 2024

**"Robotics in the Context of Large Pre-Trained Models"**
- Perception, Action, and Learning Group at UPenn (Prof. Dinesh Jayaraman) February 2024
- NTU Robot Learning Lab (Prof. Shao-Hua Sun) October 2023

**"Learning to Synthesize Programs as Interpretable and Generalizable Policies"**
- AIPlans Workshop at NeurIPS 2021 December 2021

## Selected Press Coverage

[P1] "[Physical AI] The first glimpse of Gemini's rampaging incarnation" by Donga Science Magazine about our ReWiND paper, December, 2025.

[P2] "REPLAB: A low-cost benchmark platform for robotic learning," by Ingrid Fadelli, *Tech Xplore*, May 29, 2019.

Last Update : October 18, 2026